

Problem I: Which one of CH_3^- and CF_3^+ is pyramidal and which one is planar?

Problem II: Predict the shape of thionyl dichloride.

Problem III: Predict the shape of sulphuryl (sulphonyl) dichloride.

Problem IV: Predict the shape of SnCl_3^- .

Problem VI: Predict the shape of arsonium ion.

Problem VII: Predict the shape of the borohydride ion.

Problem VII: Predict the shape of HCN molecule.

Problem VIII: Predict the shape of nitrosyl chloride.

Problem IX: Predict the shape of chlorate ion $[\text{ClO}_3]^-$.

Problem X: Predict the shape of carbonium ion $[\text{CH}_3]^+$.

Problem XI: Predict the shape of carbanion $[\text{CH}_3]^-$.

Problem XII: Explain the T-shape of ClF_3 .

Problem XIII: Explain the irregular tetrahedral shape of the POCl_3 , molecule with Cl-P-Cl bond angle being 103° .

Problem XIV: PCl_5 , has the shape of a trigonal bipyramid whereas IF_5 , has the shape of a square pyramid. Account for this difference.

Problem XV: Predict the shape of F_2O .

Problem XVI: Explain the relative bond angles (O-N-O) in the series of $[\text{NO}_2]^+$, $[\text{NO}_2]$ and NO_2^-

Problem XVII: Predict the shapes of acetylene and ethylene molecules.

Problem XVIII: Predict the shape of ClO_2^- , ion.

Problem XIX: Predict the shapes of carbonyl chloride (COCl_2) and carbonyl sulphide (COS).

Problem XX: How the oxonium ion will look like?

XXI: Predict the shape of the molecules

NCBr , BO_3^{3-} , XeOF_2 , XeF_2 , BrF_2^- , NSF , ICl_2^{2+} , IO_3^- , $[\text{IOF}_5]^-$, $[\text{SbF}_5]^{2-}$, $[\text{AlF}_6]^{3-}$, $[\text{XeF}_5]^-$, $[\text{TeF}_7]^-$, $[\text{IO}_2\text{F}_5]^{2-}$